Web Scan Pro:

📌 Project Overview

Web Scan Pro is a custom-built web application security scanner developed for educational purposes.

The project focuses on:

- Setting up vulnerable web applications locally
- Developing a crawler to scan web pages
- Extracting forms and input fields
- Structuring metadata for future security testing modules

The target application used is:

- DVWA (Damn Vulnerable Web Application)
- Hosted locally using XAMPP

🗓️ Week 1 – Project Initialization & Setup

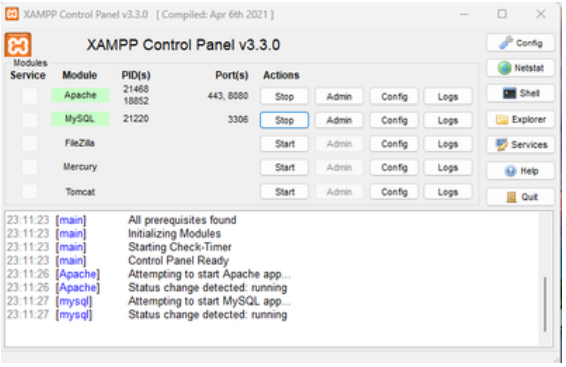
🎯 Objectives

- Define project goals and expected outcomes
- Set up DVWA locally
- Explore application structure and vulnerabilities
- Prepare environment for scanning

🛠️ Environment Setup

Install XAMPP

- Download XAMPP
- Install it on Windows
- Start Apache and MySQL



- XAMPP Control Panel showing Apache and MySQL running (green)

Install DVWA

- Download DVWA
- Place it inside:

  C:\xamppss\htdocs\

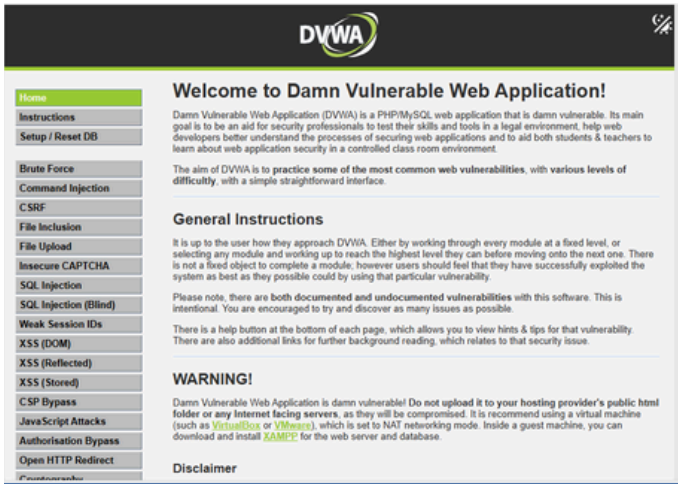
- Access in browser:

  http://localhost/dvwa

- Go to:

  http://localhost/dvwa/setup.php

- Click Create / Reset Database



Configure DVWA

Login with:

- Username: admin
- Password: password

Set Security Level to:

- LOW

- DVWA Security Level set to LOW

🔎 Application Exploration

The following were analyzed:

- Login functionality
- Form-based vulnerabilities
- URL structure
- GET and POST requests
- Input field behavior
- Hidden CSRF tokens

Outcome:

✔ Understood structure of vulnerable endpoints
✔ Identified interactive elements
✔ Prepared environment for automated scanning

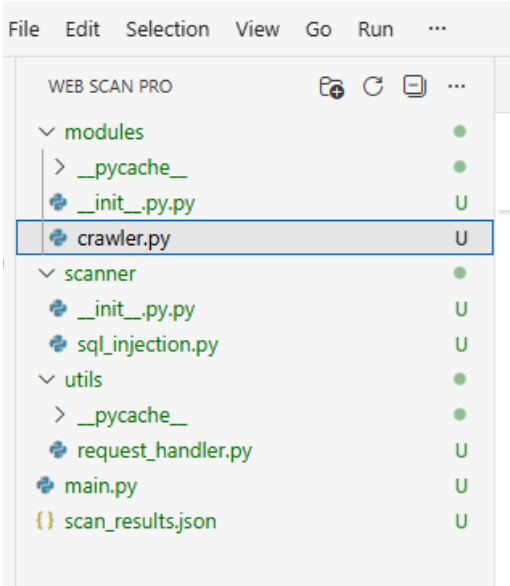
📅 Week 2 – Target Scanning Module

🎯 Objectives

- Develop a web crawler
- Discover pages and links
- Extract forms and input fields
- Structure and save metadata

🏗 Project Structure

```
Web scan pro/
│
├── main.py
├── modules/
│   ├── __init__.py
│   └── crawler.py
│
├── utils/
│   ├── __init__.py
│   └── request_handler.py
│
└── scan_results.json
```



- VS Code showing project folder structure

Request Handler Module

Responsible for:

- Managing HTTP session
- Handling login
- Extracting CSRF token
- Maintaining authentication

Key Features:

- Uses requests.Session()
- Extracts hidden user_token
- Submits login credentials programmatically

Web Crawler Module

Responsible for:

- Crawling pages recursively
- Discovering links
- Extracting forms
- Extracting input fields
- Saving structured metadata
- Technologies Used:
- BeautifulSoup
- urllib.parse
- JSON module

🔄 Crawler Workflow

- Start at base URL
- Send GET request
- Parse HTML using BeautifulSoup
- Extract all <form> elements

Extract:

- Form action
- Method (GET/POST)
- Input fields
- Store results in metadata list
- Recursively follow internal links
- Save results to scan_results.json

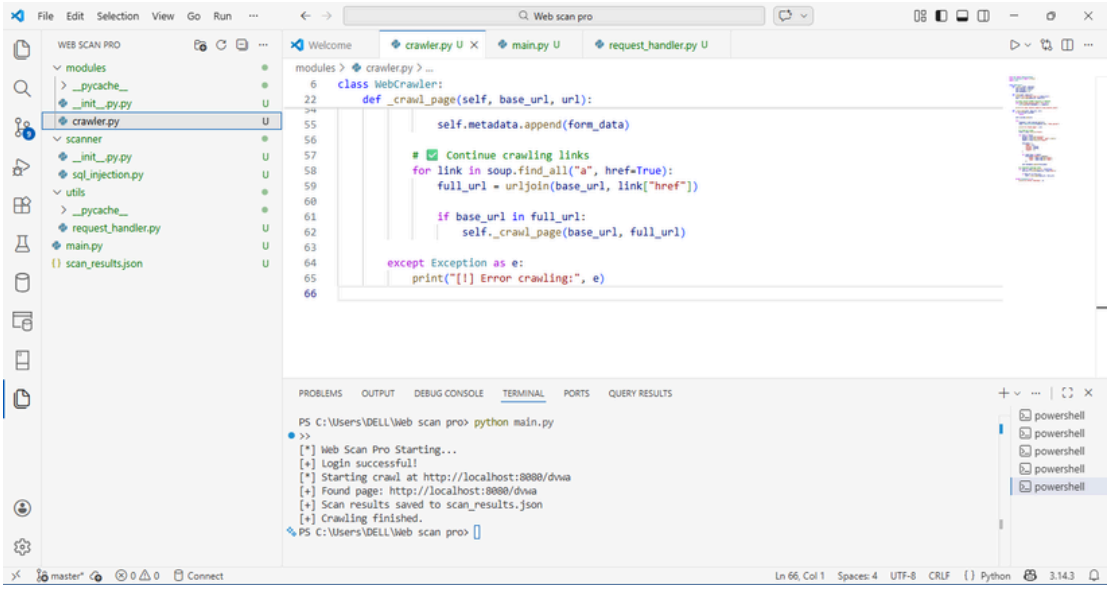
📂 Output Example

After running:

```
python main.py
```

Console Output:

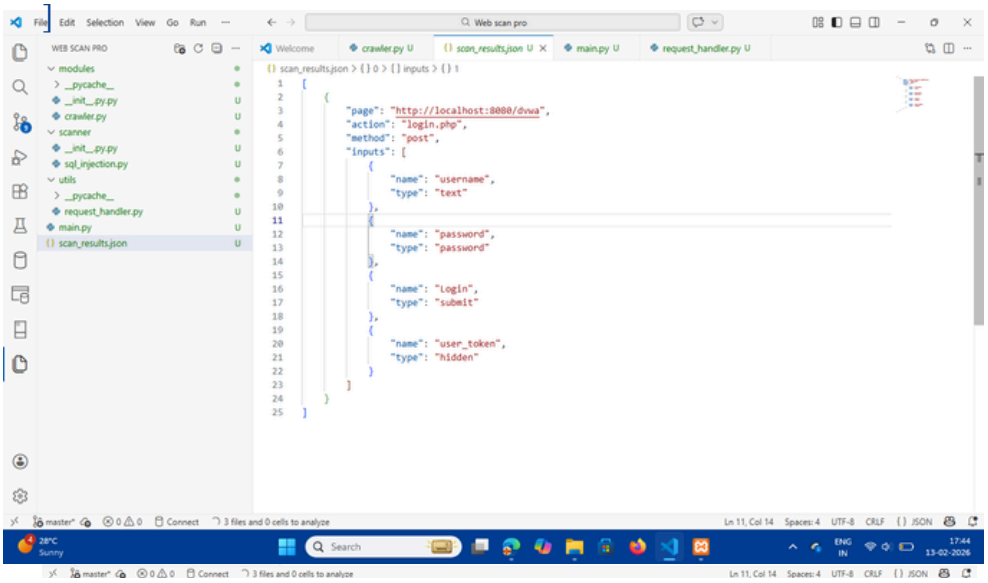
```
[*] Web Scan Pro Starting...
[+] Login successful!
[*] Starting crawl at http://localhost:8080/dvwa
[+] Found page: ...
[+] Scan results saved to scan_results.json
```



📦 JSON Output

Example structure:

```
[
  {
    "page": "http://localhost:8080/dvwa",
    "action": "login.php",
    "method": "post",
    "inputs": [
      {"name": "username", "type": "text"},
      {"name": "password", "type": "password"},
      {"name": "Login", "type": "submit"},
      {"name": "user_token", "type": "hidden"}
    ]
  }
]
```



- scan_results.json file opened in VS Code

🧪 Manual Execution Steps

1. Step 1 – Start XAMPP
- Start Apache
- Start MySQL
2. Step 2 – Ensure DVWA is Accessible

Open browser:

http://localhost/8080/dvwa

Step 3 – Run Scanner

- Open terminal inside project folder:

```
cd "C:\Users\DELL\web_scan_pro"
python main.py
```

Step 4 – Verify Output

- Check console output
- Open scan_results.json
- Confirm extracted forms and inputs

✅ Achievements

By the end of Week 2, the project successfully:

✔ Set up vulnerable target environment
✔ Built authenticated crawler
✔ Extracted interactive elements
✔ Structured metadata for testing modules
✔ Saved scan results in JSON format